

Student Engagement and Interactive Learning Framework for Smart Classes

From initial contribution to a full architecture

Presented by: **Batoul Kanaan** · Supervised by: **Dr. Hassan Harb**



Where Current Systems Fall Short

Existing solutions are fragmented - there is no unified architecture that connects content, participation and prediction end-to-end.

Fragmented Systems

Engagement tools and prediction models operate in isolation

Limited Short-Answer Analysis

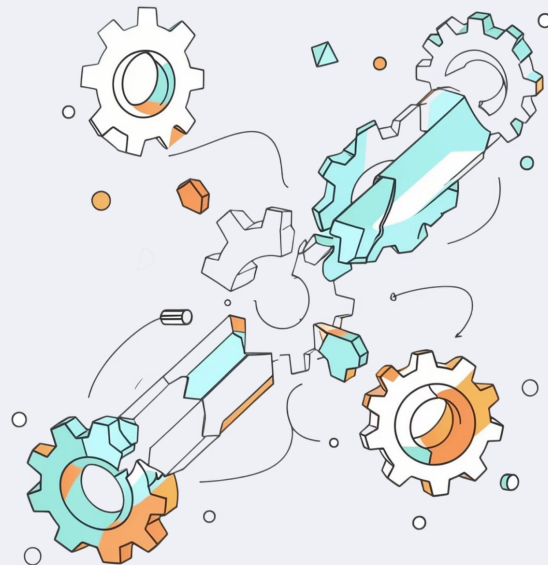
Open-ended student responses are rarely understood deeply

Disconnected Dashboards

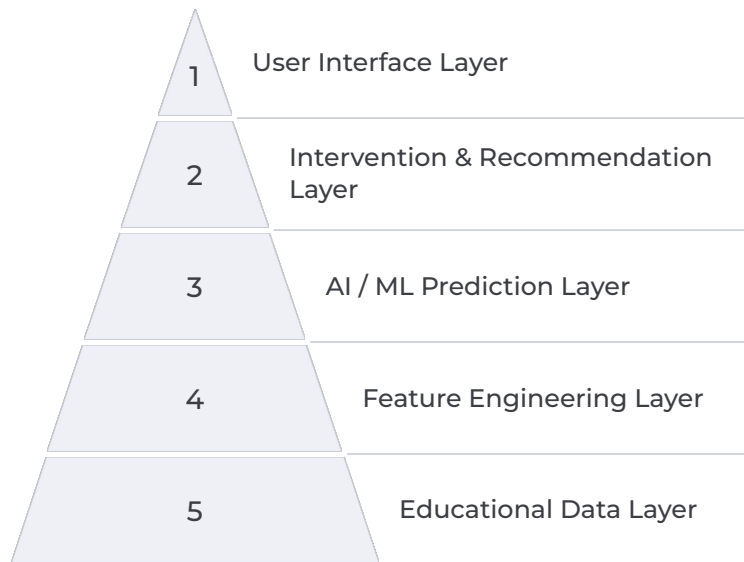
Analytics are not tied to real-time classroom participation

Missing Admin Oversight

Institutional-level monitoring is largely absent from current tools



Initial Contribution — Smart Education Analytical Framework



Core Objective

Use AI-powered analytics to turn educational interactions into smart educational decisions.

Main Components

- Engagement Analysis
- Performance Prediction
- Recommendation System
- Multi-Dimensional Engagement Modeling

Research Contribution

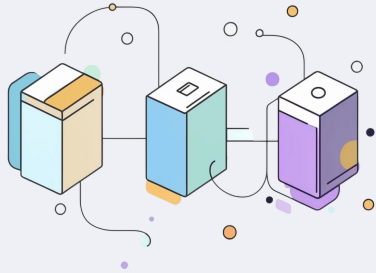
Established the conceptual foundation for integrating analytics, prediction, and personalized educational support inside one intelligent framework.

- ✓ This framework established the analytical and educational foundation that guided the development of the new operational architecture — transforming the vision into a complete system with AI content generation, real-time participation tracking, and live predictive analytics.

NEW CONTRIBUTION

Proposed Three-Stage Architecture

This contribution is not a single module but a connected end-to-end architecture unifying content generation, participation tracking, and predictive analytics.



1

Stage 1

AI Engagement Generation

Generative AI automatically create interactive questions out of lecture material

2

Stage 2

Participation Tracking

Collects real-time student responses via QR-based live sessions during class

3

Stage 3

Analysis & Prediction

Dashboards

Analyzes participation data to predict engagement and guide instructor and admin decisions



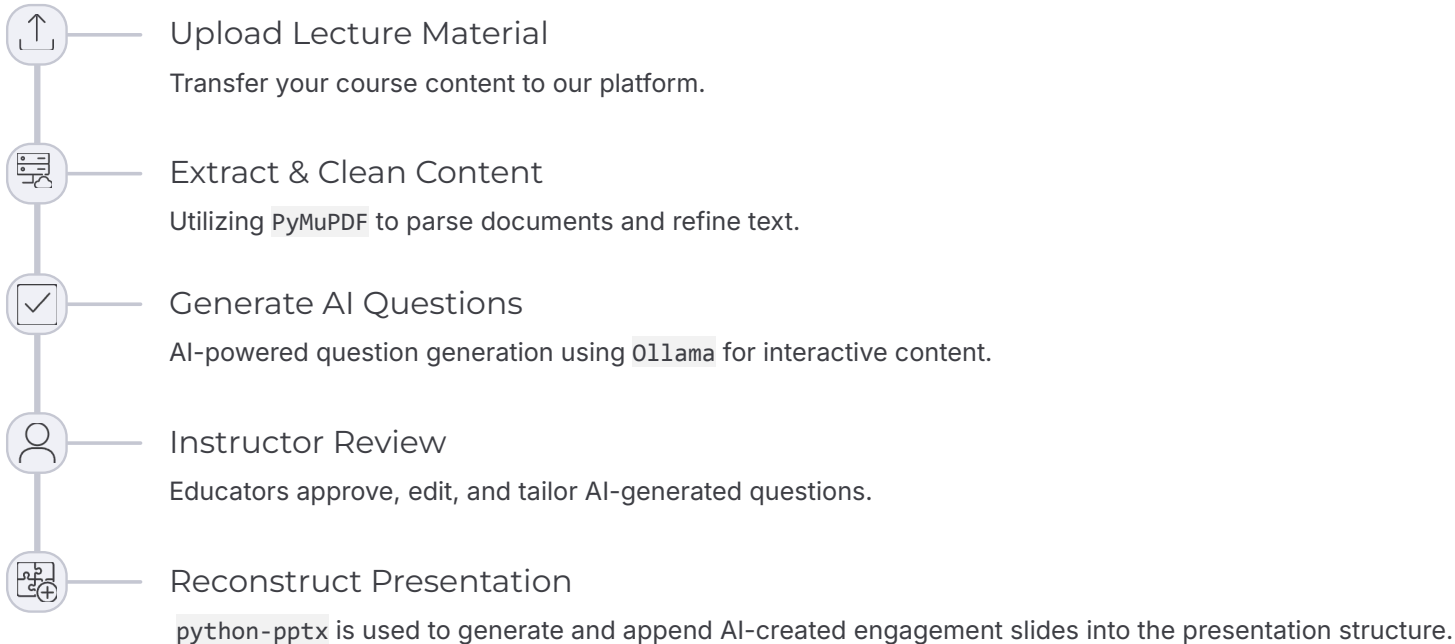
STAGE 1

AI Engagement Generation

Stage 1 automatically converts the static lecture material into an interactive classroom engagement content. Lecture files are turned into questions by AI. Instructors then review the questions and can use them right away in class.

Stage 1 Workflow

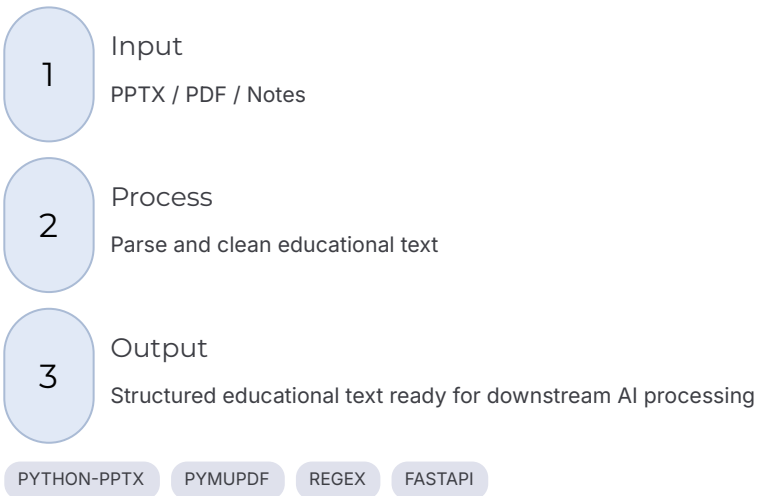
Our workflow efficiently converts your lecture material into engaging, AI-powered questions, ready for immediate use in the classroom.



Content Extraction Module

The system parses PowerPoint slides, PDF chapters, and lecture notes — extracting readable educational text while filtering out non-text elements.

The result: clean, machine-readable content ready for AI processing.



Extracted

Titles, bullet points, text boxes, table content



Ignored

Images, audio/video, unsupported visual objects



AI Question Generation Module

Cleaned lecture content is sent to a locally deployed LLM, which generates formative engagement questions — supporting privacy, security, and offline processing while encouraging active learning.



Local LLM

Ollama runs on-device for private, secure, offline question generation.



3 MCQs

Questions with distractors and correct-answer explanations.



1 Short Answer

One open-ended prompt for deeper reflection and recall.



Bloom's Tags

Difficulty labels and cognitive level markers for each item.



Answer Explanations

AI-generated rationales for every answer option.

Module Flow



Input

Cleaned lecture content



Process

LLM processing



Output

Structured questions

Technologies Used

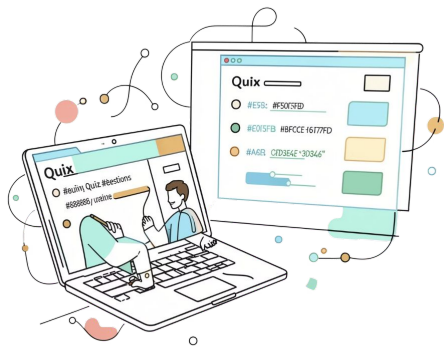
OLLAMA

LLAMA 3.1

FASTAPI

PYDANTIC

Instructor Review & Presentation Reconstruction



Before any question reaches students, the instructor reviews, edits, and approves each item. Once approved, the system automatically reconstructs a fully updated PowerPoint presentation with embedded engagement slides.

PYTHON-PPTX

REACT

FASTAPI



Review

Instructor inspects AI-generated questions in React UI



Approve

Edits and confirms questions for classroom use



Re-Generate

Updated PPTX generated automatically

Stage 1 Final Output

The result is no longer static lecture content — it becomes an interactive, classroom-ready experience powered by AI.

Before

Traditional Lecture Presentation

- Passive, one-way content delivery
- Static slides
- No engagement tracking

After

AI-Enhanced Interactive Presentation

- Embedded MCQs & Short-answer questions
- Bloom's taxonomy tags
- Difficulty labels

3

MCQs Generated

1

Short-Answer Question

100%

Instructor Approved

FASTAPI

MONGODB

PYTHON-PPTX

STAGE 2

Participation Tracking & Live Classroom Interaction

Stage 2 transforms AI-generated questions into real-time classroom participation. Students engage through authenticated QR-based sessions, while instructors receive live analytics on engagement and understanding.



Real-Time Participation



Authenticated Students



Live Analytics



Participation Recording

WEBSOCKET

MONGODB

JWT AUTHENTICATION



Stage 2 Participation Pipeline



Start Session

Instructor launches live session



QR Generated

Unique code for student access



Authenticate

Students log in via JWT



Answer Questions

Students submit responses



Store Responses

Data saved in MongoDB

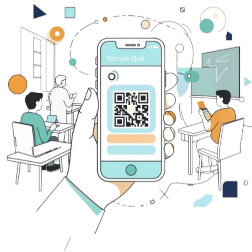


Live Dashboard

Real-time analytics update

 The system tracks both **participation** (who responded) and **correctness** (accuracy of responses) — giving instructors a complete picture of classroom understanding.

Student Authentication & Live Participation



Students join sessions through authenticated accounts — eliminating anonymous or fake participation. Every submitted response is linked to an authenticated student identity — making participation data reliable and educationally meaningful.

JWT

FASTAPI

MONGODB

REACT



QR Access

Scan to join the active session



Student Login

Verified account access



JWT Token Generated

Secure session credential issued



Authenticated Participation

Student responses tied to identity



Response Stored

Participation data saved for analysis



Input

Student session request



Process

Authentication and participation validation

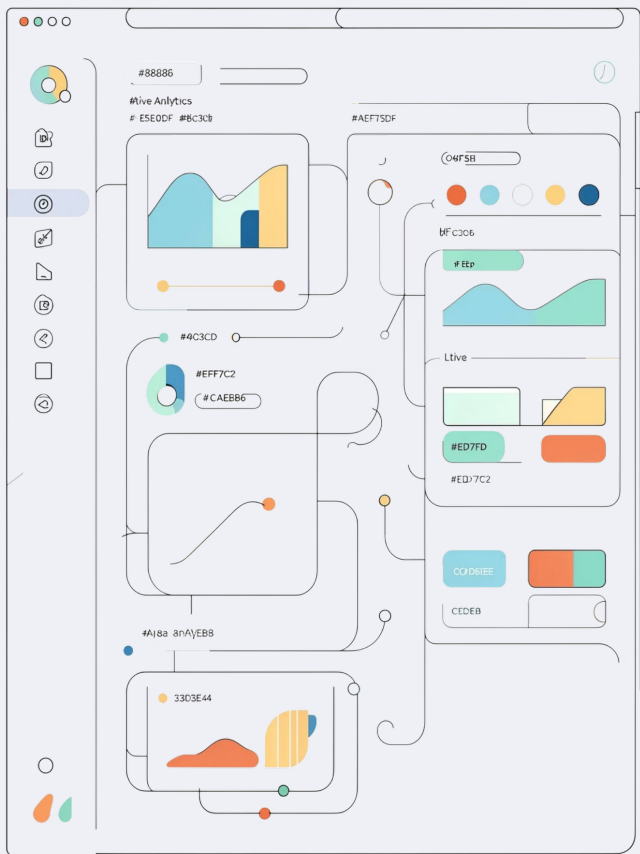


Output

Verified participation records



JWT tokens secure classroom participation after QR-based session access.



STAGE 2 · DASHBOARD

Instructor Live Dashboard

Student Responses



WebSocket Updates

Live Instructor Dashboard



MongoDB Aggregation



Participation Rate

Percentage of students who have responded



Answer Distribution

Breakdown of selected MCQ options



Correct / Incorrect

Accuracy overview across the class



Unanswered Students

Identify who hasn't responded yet



Response Activity

Live feed of incoming submissions

REACT

WEBSOCKET

MONGODB AGGREGATION

CHART.JS

 Dashboard updates are pushed via WebSocket — no page refresh required.

STAGE 3

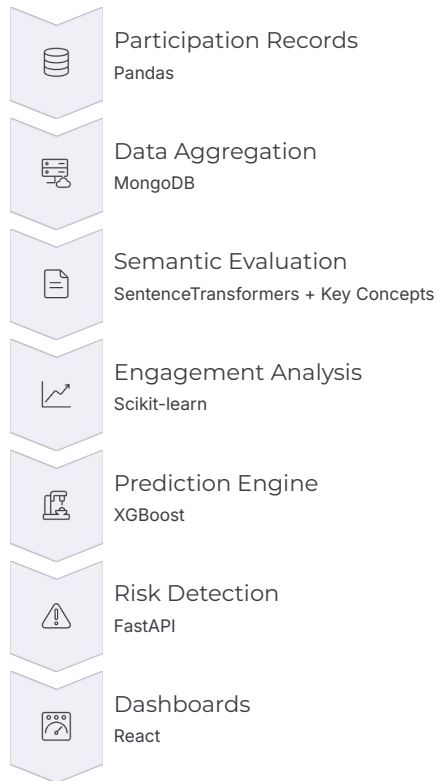
Analysis, Prediction & Educational Intelligence

After collecting classroom participation data, the system analyzes engagement patterns and predicts student performance — transforming raw interaction into actionable educational intelligence.



Stage 3 Workflow

A seven-stage analytics pipeline transforms raw classroom data into predictive educational insights — from participation records to real-time dashboards.



MODULE 1

Data Aggregation Module

The system collects and organizes participation and performance records from MongoDB, transforming raw classroom interactions into structured educational datasets ready for analysis.



MongoDB Aggregation

Pandas

Numpy

FastAPI



Participation Records

Raw responses, timestamps, correctness, and session data



MongoDB Aggregation

Aggregation pipelines compute metrics and statistical summaries



Structured Analytics Dataset

Clean, aggregated, and analysis-ready educational datasets



INPUT

Participation and performance records from MongoDB

Behavioral Metrics

- Participation Frequency: How often students engage per session
- Attendance Consistency: Session participation patterns



OUTPUT

Analytics-ready educational datasets for downstream tasks

Performance Metrics

- Answer Correctness: Accuracy rate across assessments
- Response Time: Speed of student interaction

Short-Answer Semantic Evaluation

Instead of rigid keyword matching, the system uses semantic similarity to evaluate open-ended student responses — **understanding meaning, not just words.**

$$\text{Final Score} = 0.7(\text{Semantic Similarity}) + 0.3(\text{Concept Coverage})$$

SentenceTransformers

Cosine Similarity

paraphrase-multilingual-MiniLM-L12-v2

**Student Answer**

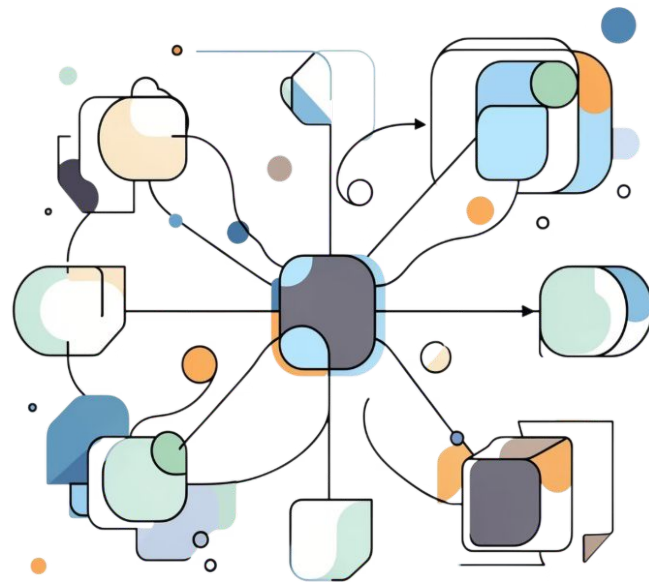
Raw open-ended text response

**Embedding Generation**

MiniLM encodes semantic meaning into vectors

**Hybrid Semantic Score**Between Semantic Similarity Scoring and Key
Concept Matching**Correct / Partial / Incorrect**

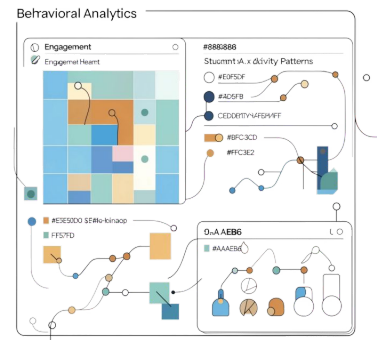
Three-tier classification of student understanding



Engagement Analysis Module

GENERATED INSIGHTS

- Engagement Score
- Participation Trends
- Low-Engagement Detection
- Behavioral Consistency



Participation Records

Raw session interaction data



Feature Extraction

Behavioral signals isolated per student



Engagement Metrics

Quantified participation and consistency scores



Behavioral Insights

Actionable intelligence for prediction models

Participation Rate

Frequency of active responses per session

Interaction Consistency

Steadiness of engagement over time

Response Timing & Quality

Patterns in answer speed and correctness

Activity Patterns

Temporal engagement trends per class

Attendance

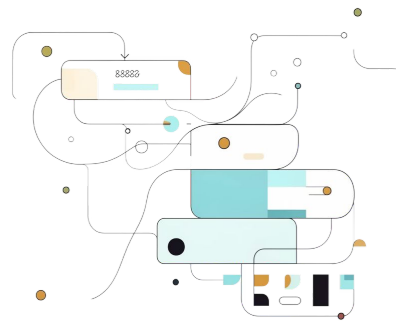
Session presence and consistency

MODULE 4

Performance Prediction Module

INPUT FEATURES

- Participation Frequency
- Attendance Consistency
- Semantic Evaluation Score
- Response Time
- Response Quality



The model learns behavioral and performance patterns to forecast future academic outcomes.



Engagement Features

Behavioral and performance signals



Feature Vector Construction

Structured input for ML model



XGBoost Prediction Model

Trained on historical student data



Risk & Performance Forecasts

Actionable predictions per student

Predicted Academic Performance

Forecasted grade trajectory per student

Engagement Drop Detection

Early warning of dropping participation

Academic Risk Scoring

Likelihood of academic underperformance

XGBoost

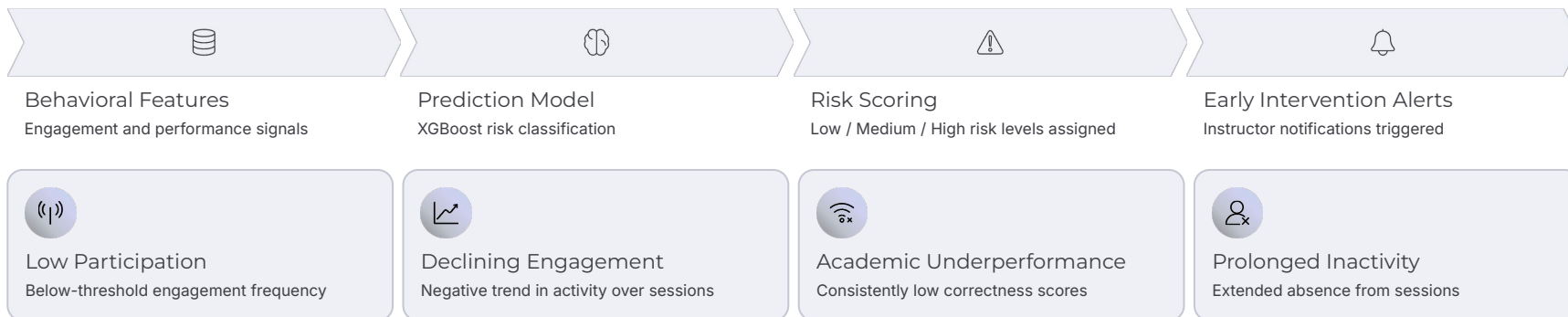
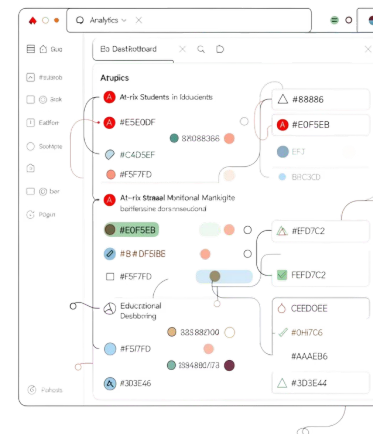
Numpy

Risk Detection & Early Intervention

INTERVENTION ACTIONS

- Instructor Alerts
- Engagement Recommendations
- Targeted Academic Support
- Student Follow-Up

RISK LEVELS



Instructor Analytics Dashboard



Participation Data

Raw session interactions and responses



Analytics Engine

Aggregation, scoring and ML predictions



Instructor Insights

Actionable classroom intelligence



Engagement Scores

Per-student and class-wide metrics

Low-Performance Topics

Topics with low correctness rates

At-Risk Students

Flagged learners needing support

Answer Distributions

Response patterns per question

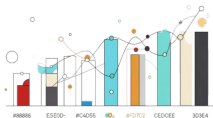
Participation Trends

Longitudinal engagement evolution

Real-time and post-session analytics empower instructors with the intelligence needed to personalize teaching and support every student effectively.

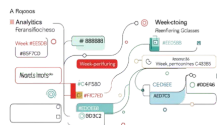
Administration / Principal Dashboard

Institution-wide monitoring provides a bird's-eye view of educational health for administrators, from engagement trends across classes to performance prediction reports.



Engagement Trends

Cross-class participation and activity comparison



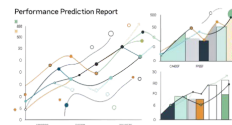
Weak-Performing Classes

Automated identification of underperforming groups



At-Risk Student Groups

Aggregated risk indicators by cohort



Prediction Reports

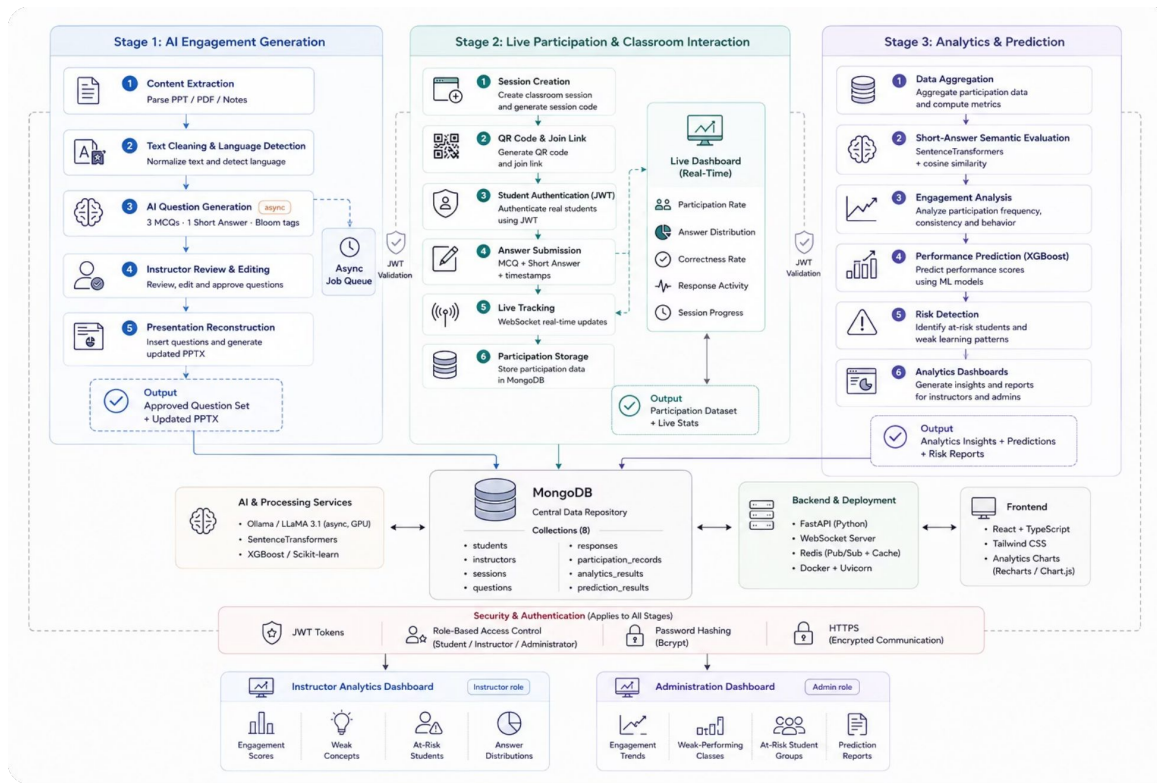
ML-driven performance forecasts institution-wide

Technologies: React · MongoDB Aggregation · FastAPI · Recharts

FULL ARCHITECTURE

Full AI-Powered Educational Architecture

A unified, intelligent ecosystem integrating content generation, real-time classroom interaction, and advanced learning analytics to transform education.



Evaluation Datasets

PRIMARY EVALUATION DATASET

OULAD Dataset

(Open University Learning Analytics Dataset)

Contains

Student interaction logs
Assessment records
Engagement activity
Final outcomes

Used For

Engagement analysis
Performance prediction
At-risk student detection

SECONDARY BENCHMARK DATASET

KDD Cup 2015 Dataset

(MOOC Learning Behavior Dataset)

Contains

Clickstream activity logs
Temporal learning behavior
Dropout-related patterns
Large-scale MOOC interactions

Used For

Behavioral analytics
Risk prediction
Engagement trend analysis



Why Both Datasets?

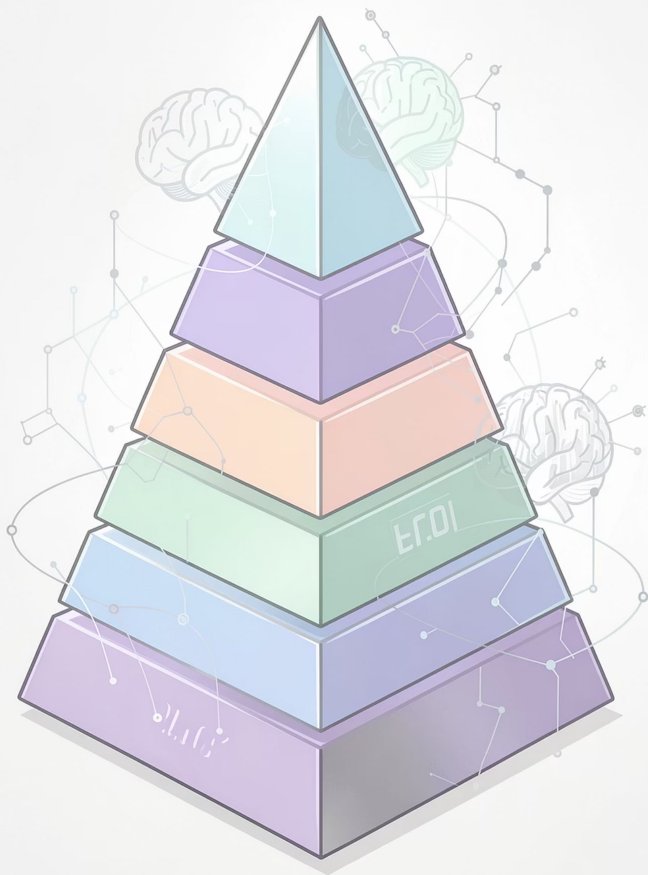
Using both datasets enables evaluation of the engagement analytics, prediction, and risk detection components across different educational environments.



An illustration of a modern office meeting. A man in a grey blazer stands on the left, gesturing towards a large screen displaying various charts and graphs. Five people are seated around a white oval table, looking at the screen or their own devices. Another man in a grey blazer stands on the right, leaning over the table. The background shows a city skyline through large glass windows.

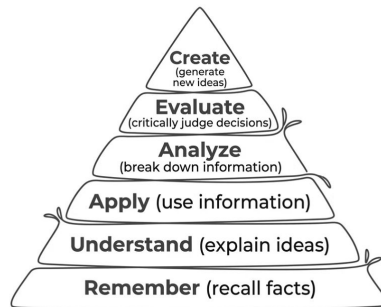
Thank You!

We look forward to transforming education together.

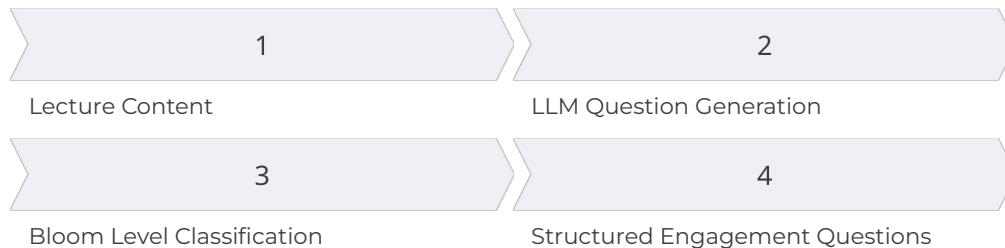


Bloom's Taxonomy in AI Question Generation

Our system moves beyond random question generation. Each AI-generated question is meticulously aligned with a specific cognitive learning level from Bloom's Taxonomy, ensuring depth and relevance.



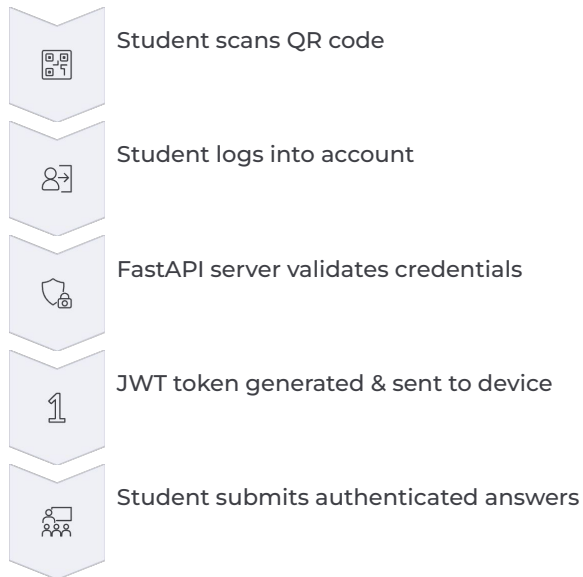
Architecture Connection



JWT Authentication in Live Classroom Participation

How secure student participation works inside Stage 2

Simple JWT Flow Diagram



Understanding JWT in our System

What is JWT?

JWT (JSON Web Token) is a secure digital token used to verify student identity during classroom participation.

Why is it used?

- Prevent anonymous participation
- Ensure each response belongs to a real student
- Secure classroom sessions
- Maintain reliable analytics data

Architecture Role

JWT connects authentication with live participation tracking, making engagement analytics trustworthy.

Hybrid Semantic Evaluation for Short Answers

Student Answer Pipeline



Why Hybrid Evaluation?

Semantic Understanding

Understands meaning, not exact words

Concept Verification

Checks essential learning concepts

Fast Automated Evaluation

Supports real-time classroom analytics

Better Educational Reliability

More accurate than keyword matching alone

Stage 1: LLM generates: short-answer question, expected answer, key concepts ↓ feeds into Stage 3: Semantic Evaluation Module → Analytics & Prediction Engine

The proposed hybrid semantic evaluation combines embedding similarity with concept coverage to provide explainable and scalable automated grading for classroom short-answer assessment.